COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. **(L2)**

Assessment Tool: Code along exercise: Make your first HTML page together with CSS

Weightage: 10%

QUESTIONS: **Total Marks: 10**

Scenario: Web Development Standards Implementation

A start up is developing a modern web application. During development, the team encounters multiple issues related to HTML elements, HTTP methods, XML syntax, and web communication protocols.

The following observations are made:

- The team needs to embed a promotional video on the homepage.
- A registration form must securely send user data to the server.
- The navigation menu must follow semantic HTML standards.
- XML configuration files are being used but contain syntax errors.
- The application must communicate over the correct web protocol.

Data Snippets Provided

Snippet 1: Video Embedding Options

A. <media src="video.mp4"></media>
B. <video src="video.mp4" controls></video>
C. <movie src="video.mp4"></movie>
D. <vid src="video.mp4"></vid>

Snippet 2: Form Submission Methods

A. GET
B. POST
C. PUT
D. DELETE

Snippet 3: Navigation Elements

A.
B. <nav>
C. <section>
D. <header>

Snippet 4: XML Syntax

A. <note><to>User</to><from>Admin</from>
B. <note><to>User</to><from>Admin</from></note>
C. <note><to>User</from></note>
D. <note to="User"></note

Snippet 5: Web Communication Protocols

A. FTP
B. HTTP
C. SMTP
D. SNMP

Questions:

1. Select the **correct HTML tag for video embedding** and justify your answer.
2. Identify the **appropriate HTTP method for form submission** and explain why.
3. Choose the **proper semantic element for navigation** and justify its usage.
4. Identify the **correct XML syntax** and explain the error in other options.

Choose the **correct protocol for web communication** and justify your choice.
Questions and Answers

1. Select the correct HTML tag for video embedding and justify your answer.

Answer: Option B — <video>

<video src="video.mp4" controls></video>

Justification: The HTML <video> element is used to embed video content on a webpage. The controls attribute provides playback options such as play, pause, and volume control.

2. Identify the appropriate HTTP method for form submission and explain why.

Answer: Option B — POST

Justification: The POST method sends form data to the server in the request body, making it suitable for submitting registration details. For secure transmission of sensitive information, HTTPS must be used. POST alone does not encrypt data.

3. Choose the proper semantic element for navigation and justify its usage.

Answer: Option B — <nav>

<nav>

  Home

  About

  Contact

</nav>

Justification: The <nav> element defines a section containing major navigation links. It improves the semantic structure and accessibility of a webpage.

4. Identify the correct XML syntax and explain the error in other options.

Answer: Option B

<note>

  <to>User</to>

  <from>Admin</from>

</note>

Justification: Option B is correct because all elements are properly opened and closed, nested correctly, and enclosed within a single root element.

Errors in the other options:

- **Option A:** The root element <note> is not closed.
- **Option C:** The opening tag <to> is incorrectly closed with </from>, resulting in mismatched tags.
- **Option D:** The closing angle bracket > is missing from the closing tag </note.

5. Choose the correct protocol for web communication and justify your choice.

Answer: Option B — HTTP

Justification: HTTP (Hypertext Transfer Protocol) enables communication between web browsers and web servers to request and deliver web content. HTTPS is the secure version of HTTP and is recommended for modern applications that handle sensitive information.

Code of Conduct:

I B.Abhsihek 192311175 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.